MATH329 · OPERATIONS RESEARCH

What repeats in the midterm

An analysis of four midterms: 472, 471, 462 and 461

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What's inside

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01 The shape of the midterm

The instructor repeats the same template every term: a short theory question, then a graph, then standard form, then simplex. The time is 60 minutes.

Question	472	471	462	461
Theory	T/F · 7	T/F · 5	list · 5	blanks · 4
Graphical	8	9	7	6
Standard form with an unrestricted variable	—	—	8	6
Standard form + Simplex	5 + 10	✓	10	9
Total	**30**	—	**30**	**25**

The 471 paper reached me without its header and the marks of its last question are not shown, so no total is given.

The bottom line

The graph, the standard form and simplex appeared in all four midterms and carry more than three quarters of the marks: 23 of 30 in 472, 25 of 30 in 462, and 21 of 25 in 461.

02 The theory question: everything that appeared

Short answers and blanks

Question	Answer	Where
The formal activities of OR were initiated in _____ during World War II.	**England**	461
An optimization problem is called _____ when the objective function and all the constraints are linear functions of the decision variables.	**LPP**	461
What does the Standard Form mean?	1. All decision variables are nonnegative. 2. All constraints are equations with nonnegative RHS.	461
Mention the special cases when solving graphically.	1. Alternative optimal solutions 2. Unbounded solution 3. No feasible solution	462
What are the two requirements to start the simplex algorithm?	1. The coefficients of the basic variables form an identity matrix. 2. Their coefficients in the objective function row are zero.	462
What methods can be applied to solve the given LPP? (two variables)	Graphical method and Algebraic method.	462 · 461

True / False

Statement		Where
The solution of the LPP may be compatible with its constraints imposed by the available resources.	**F**	472 · 471
Graphical solutions are only applicable to LP problems with more than three variables.	**F**	472
Operations Research was first initiated during World War I in England.	**F**	472
In a linear programming problem, the objective function or the constraints must be linear.	**F**	472
The Simplex tableau is used to organize the objective function and constraint equations for the algorithm.	**T**	472
The Simplex Algorithm requires the coefficients of the starting basic variables to form an identity matrix.	**T**	472
All the parameters of the model are unknown.	**F**	471
A basic feasible solution is a feasible solution to LPP which is also the basic solution.	**T**	471
A basic feasible solution is called degenerate if value of at least one basic variable is zero.	**T**	471
If all the ratios in simplex table are negative or infinite then the solution an alternative.	**F**	471

Theory questions rarely repeat word for word, but they all come from the same few sentences on the slides. The instructor likes to change one word: **II** becomes **I**, **and** becomes **or**, **must** becomes **may**, **unbounded** becomes **alternative**.

A seventh 472 item about "no variable can enter the basis" is left out here, because the key's answer to it is doubtful. The details are in the 472 solutions file.

03 The graphical question

Midterm	Problem	The twist	Answer
472	Max $5x_1 + 3x_2$	The constraint $-2x_2 \geq -4$, and a fractional answer.	25.5 at (4.5, 1)
471	Max $3x_1 + 2x_2$	The constraint $-x_1 \geq -4$.	18 at (2, 6)
462	Max $3x_1 + 2x_2$	The constraint $x_2 \geq 1$, a negative coefficient $-2x_1$, and a fractional answer.	7.5 at (0.5, 3)
461	Max $10x_1 + 5x_2$	A $\geq$ constraint that pushes the region away from the origin.	55 at (4, 3)

What never changes	What to prepare for
Always max, always three constraints.	One constraint has a twist: a negative right-hand side or a $\geq$ sign.
The optimum is where two lines meet.	Solve the two equations, because the answer is a fraction half the time.
There is a mark for the objective line **Obj**.	Draw it and label it.

04 The standard form question

In 461 and 462 it was a separate question with exactly the same template; only the numbers change:

Max or Min Z = a x_1 ± b x_2
S.to (≤ constraint)
(≥ constraint)
(= constraint with a negative right-hand side)
x_1 ≥ 0, x_2 unrestricted in sign

So the question combines every conversion rule: slack, surplus, multiplying by `−1` and the substitution `x₂ = x₂⁺ − x₂⁻`. It is always preceded by "What methods can be applied?", answered with **Graphical method and Algebraic method**.

In 471 and 472 this question did not appear, and the standard form became part (a) of the simplex question, with no unrestricted variable.

05 The simplex question

Midterm	Problem	Required
472	Max $16x_1 + 15x_2$ · 3 constraints	Two tables only.
471	Max $2x_1 + 3x_2$ · 2 constraints	Full solution, three tables.
462	Min $x_1 - 2x_2$ · 3 variables	Full solution, three tables.
461	Max $8x_1 + 7x_2$ · 3 constraints	Two tables only.

- All constraints are `≤` after fixing the right-hand side, so only slack variables are needed.

- The 472 question is a copy of the 461 question with different numbers: the same constraints, the same entering x_1 and leaving s_3.
- Min came once (462), so remember that its entering variable is the most positive entry.
- It is marked by row: each row carries a mark and each arithmetic slip costs half a mark.

06 What never appeared

Topic	Note
A word problem to formulate	None in the four. Every problem came with its equations ready.
The table of basic solutions	Never as a problem. Only the definition of a BFS appeared in true/false.
Special cases with tableaus	Never solved. Only their definitions appeared in theory.
Example 3.5 (absolute values)	Never appeared.
Big-M	Only in quiz 3-4 of term 471, never in a midterm.

Caution

Four midterms are a small sample. "Never appeared" does not mean "will not appear", especially formulation, since the instructor taught many examples of it. Study it after mastering the fixed questions.

07 Study order

#	What to study	File
1	The graphical method, with a constraint that has a twist.	Lecture 2
2	Simplex: two tables, a full solution, and a min problem.	Lecture 4
3	Standard form with an unrestricted variable and a negative right-hand side.	Lecture 3
4	The theory sentences in the two tables above.	This file
5	Solve the four midterms and the mock exam against the clock.	Midterms and practice sections
6	Formulation and basic solutions.	Lecture 1 · Lecture 3